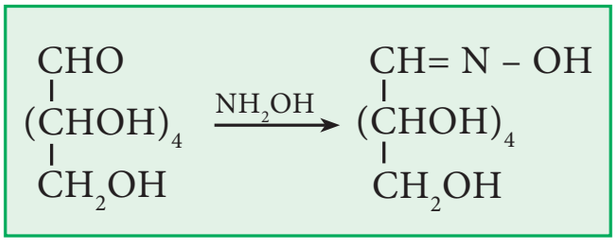
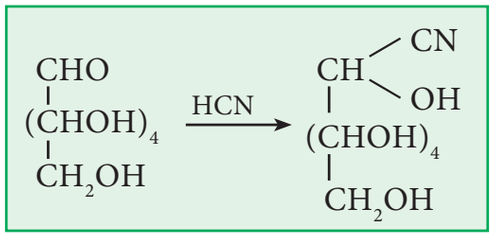


x. Action with HCN (Kiliani synthesis):

Both glucose and fructose form cyanohydrins with HCN.



Note

Tollen's Reagent:

Ammonical Silver Nitrate ($[Ag(NH_3)_2]NO_3$)

Fehling's Reagent:

A- Copper(II) Sulfate

B- aqueous Potassium sodium tartrate

Benedict's Reagent:

Mixture of sodium carbonate, sodium citrate and copper (II) sulfate

xi. Action with Conc. HCl:

Both glucose and fructose when heated with conc. HCl give laevulic acid.

xii. Action with alkalies:

When warmed with conc. alkali sugars first turn yellow, then brown and finally resinify. But in the presence of dilute. alkali glucose and fructose give a mixture of D-glucose, D-mannose and D-fructose. This is known as Lobry de Bruyn - van Ekenstein rearrangement. This occurs through enediol.

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